

Article

Video Anomaly Detection Through Spatial–Temporal Feature Relocalization and Calibrated Trajectory Modeling

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Abstract

To address the limitations of existing video anomaly detection methods that overly rely on pixel-space reconstruction and are sensitive to background noise and object scale variations, a self-supervised contrastive learning approach that integrates spatial–temporal feature relocalization with camera-calibrated trajectory modeling is proposed. The proposed method takes spatial–temporal feature relocalization as the core task and constructs a feature-level contrastive learning mechanism to guide the model to focus on discriminative local appearance variations and global temporal semantic evolution. While suppressing background interference and scale-related noise, the method enhances the modeling of fine-grained appearance anomalies and global action-related temporal anomalies. Furthermore, camera calibration is introduced to recover continuous object trajectories in physical space, and a temporal aggregation module is designed to jointly model object motion patterns in pixel space and physical space, thereby improving the model’s ability to perceive complex anomalous behaviors. Experimental results on multiple public video anomaly detection benchmarks demonstrate that the proposed method consistently outperforms existing approaches, validating its effectiveness and generalization capability.

Keywords: video anomaly detection; self-supervised learning; camera calibration; contrastive learning

1. Introduction

Video Anomaly Detection (VAD) aims to automatically detect events or objects that deviate from normal behavior patterns, and plays a critical role in applications such as public security, traffic surveillance, and emergency response. Due to rarity, open-endedness, and the difficulty of exhaustive enumeration, comprehensive annotation of anomalous events becomes infeasible in real-world settings. Accordingly, video anomaly detection is commonly formulated as an out-of-distribution (OOD) detection problem under unsupervised or self-supervised learning strategies, where models are trained solely on normal data to learn a compact representation of regular spatial–temporal patterns, and anomalies are detected by measuring deviations from this learned representation.

Early video anomaly detection methods predominantly rely on handcrafted or shallow features to model normality, which are inherently limited by insufficient representational capacity and poor scene adaptability. With the rapid advancement of deep learning, reconstruction- and prediction-based approaches that exploit appearance and motion cues [1,2] have gradually become widely adopted. These methods are typically developed



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under an unsupervised setting, in which models are trained on normal video data and anomalies are detected by evaluating deviations in reconstruction [3–8] or prediction [9,10] outcomes. DecoAD [11] proposes a novel decoupling-based anomaly detection architecture to disentangle and interweave scene and action representations, thereby enabling explicit modeling of their complex relationships.

Despite their success in standard benchmarks, existing approaches largely base anomaly discrimination on reconstruction errors in pixel space or low-level feature space, which makes them sensitive to background clutter, local appearance variations, and object scale changes. To address these limitations, some studies [9,12] introduce multi-scale feature extraction architectures to enhance the model's capability in capturing scale variations of target objects. Zhang et al. [13] further propose a self-supervised multi-task learning framework that explicitly guides the model to focus on foreground motion dynamics and performs anomaly scoring in the feature space, thereby alleviating the adverse effects of background noise and scale bias on anomaly discrimination to a certain extent.

However, existing approaches primarily formulate anomaly detection objectives around pixel-wise or feature-wise reconstructability, rather than the semantic coherence of spatial–temporal representations. Such a formulation intrinsically entangles abnormal cues with background clutter and scale-induced variations, making it difficult to fundamentally eliminate the influence of background noise and scale bias. Reconstructability-based criteria further exhibit inherent limitations in modeling scale variations. Frame-level video anomaly detection suffers from scale noise, as reconstruction difficulty varies across objects of different spatial extents. Importantly, object-level modeling does not fundamentally resolve this issue. Normalizing objects of varying original scales to a fixed resolution introduces discrepancies in semantic density, thereby leading to implicit scale noise.

Furthermore, most existing video anomaly detection methods model motion primarily in pixel space, with limited attention to object motion behaviors in the physical space. The motion pattern of the same object exhibits intrinsic consistency across pixel and physical spaces. In complex behavior analysis, motion trajectories in the physical space provide more stable and discriminative behavioral constraints for pixel-level feature learning.

To address the aforementioned limitations, we propose a self-supervised video anomaly detection framework that integrates spatial–temporal feature relocalization with calibrated trajectory-based motion modeling. To mitigate background clutter and scale noise, normal behaviors are modeled by enforcing semantic consistency of features across spatial and temporal dimensions, leading to a self-supervised anomaly detection strategy based on spatial–temporal relocalization. Specifically, feature-level relocalization constraints are constructed independently along the spatial and temporal dimensions, guiding the model to learn stable spatial correspondences of local appearance structures and consistent temporal evolution of global representations. This design enables anomaly discrimination to focus on the plausibility of behavior patterns themselves, rather than reconstructability.

To capture physical motion dynamics beyond pixel-space representations, we further introduce a camera calibration approach that does not require explicit intrinsic or extrinsic parameters as input, enabling the recovery of object trajectories in the physical space. A multimodal image–trajectory modeling module is then employed to jointly capture the consistency between pixel-space projections and physical-space motion patterns. Extensive experiments on multiple public video anomaly detection benchmarks demonstrate that the proposed method consistently outperforms existing approaches in anomaly detection performance, validating its effectiveness.

The contributions of our work are summarized as follows:

- We propose a feature-level spatial–temporal relocalization contrastive learning task to replace pixel-level reconstruction or prediction, aiming to directly model the semantic plausibility of spatial–temporal patterns.
- We introduce a calibration-free camera calibration module that maps object motion from pixel space to physical space, and design a multimodal fusion module for collaborative modeling.
- Extensive experiments on multiple standard datasets demonstrate that our method achieves state-of-the-art performance, and ablation studies further validate the effectiveness of each proposed module.

2. Related Work

2.1. Appearance- and Motion-Based Video Anomaly Detection

Appearance- and motion-based video anomaly detection methods often adopt an unsupervised paradigm, treating pixel-level reconstruction or prediction as the core task and using reconstruction or prediction deviations to detect anomalies. Reconstruction-based approaches use autoencoders [3–5], or GANs [6–8] to reconstruct image frames or motion fields. Large reconstruction errors are treated as indicators of anomalies. Prediction-based approaches [9,10] focus on modeling temporal evolution by forecasting future frames, optical flow, or spatial–temporal features. When prediction diverges from actual observation, the discrepancy is used as an anomaly signal. To tackle the challenges posed by multi-scale behaviors in complex scenes, some studies introduce multi-scale modeling mechanisms to enhance the representation of objects at different scales. For instance, Wang et al. [9] design various pooling operations to extract image features at multiple resolutions for prediction, effectively capturing normal behavior patterns of objects across scales; Zhong et al. [12] leverage a feature pyramid structure to extract spatial–temporal features at multiple scale levels, facilitating the modeling of multi-scale behavior distributions in complex scenarios. Despite these efforts, such methods still predominantly rely on pixel-level reconstruction or prediction errors and do not explicitly model the semantic consistency of behaviors in spatial–temporal space, leaving them susceptible to background clutter and variations in object scale.

2.2. Self-Supervised Video Anomaly Detection

In contrast to the implicit generation tasks used in unsupervised reconstruction, self-supervised VAD methods focus on explicit proxy tasks for discriminative feature extraction from unlabeled data. For instance, Wang et al. [14] propose an instance discrimination task that clusters normal samples and identifies deviations from cluster centers as anomalies; Georgescu et al. [15] design a multi-task framework combining sequence order verification, motion regularity modeling, and intermediate-frame reconstruction to enhance the understanding of video temporal structure; Wang et al. [16] introduce a spatial–temporal jigsaw task, rearranging video frames and pixels across space and time to learn fine-grained spatial–temporal behavior patterns. While effective in capturing pre-defined perturbations, these methods primarily emphasize task-specific discrimination or reconstruction rather than explicitly modeling the intrinsic spatial–temporal regularities of normal behaviors, limiting their generalization in complex scenarios. To address this, we propose a spatial–temporal feature relocation task and a temporal image–trajectory multimodal induction task. By strengthening pixel-level behavioral understanding and incorporating physical-space motion constraints, our method reinforces the intrinsic link between physical actions and their pixel-based representations.

2.3. Camera Calibration

Camera calibration is the process of estimating a camera's intrinsic and extrinsic parameters to establish an accurate projection from the image coordinate system to the real-world coordinate system. Conventional camera calibration approaches generally employ methods such as checkerboard-based calibration [17], vanishing point estimation [18], and structure from motion [19]. Camera calibration methods based on deep learning primarily fall into three categories: pixel regression, geometric semantic learning, and image pixel-space mapping relationship learning. Pixel regression approaches [20,21] formulate camera calibration as a problem of directly regressing the intrinsic and extrinsic parameters from images. DeepVP [22] is the first deep learning-based method for vanishing point detection, which estimates camera parameters from a single image by detecting vanishing points along candidate horizons, scoring them, and exploiting geometric constraints. The CTRL-C method [23] employs a multi-task learning framework to jointly learn image pixel and geometric information for camera parameter estimation. The DSAC* model [24] introduces an end-to-end neural network that reconstructs the real-world positions corresponding to each pixel, learning the correspondence between predicted real-world coordinates and image features to estimate the camera projection matrix. However, these methods often lack real-world scale cues or rely heavily on specific training scenes, making cross-scene calibration challenging. In this work, we leverage geometric cues together with image depth information to estimate the camera pose, enabling calibration without requiring prior intrinsic or extrinsic parameters.

3. Method

This section provides a detailed description of the proposed self-supervised video anomaly detection framework that integrates spatial-temporal feature relocalization with calibrated trajectory-based motion modeling. The framework consists of three main components: self-supervised spatio-temporal feature relocalization, camera calibration, and multimodal image-trajectory behavior co-induction. Specifically, the spatial-temporal feature relocalization module is designed to enhance the model's discriminative understanding of spatial-temporal features. The camera calibration module estimates the geometric parameters of the camera from image information, thereby recovering object motion trajectories in the real-world scene. The multimodal collaborative modeling module further jointly exploits image features and calibrated trajectory information to capture the consistency between object projections in pixel space and their physical motion in real space. The overall architecture is illustrated in Figure 1.

3.1. Self-Supervised Spatial-Temporal Feature Relocalization

In this section, we design a self-supervised spatio-temporal feature relocalization task for video anomaly detection. Based on this task, a relative spatial encoding module is introduced to dynamically learn feature-level spatial relationships while preventing self-supervised label leakage caused by absolute positional information. To enhance feature discriminability and temporal stability in scenarios with subtle motion, we introduce learnable query vectors for each spatial location and each temporal frame along the spatial and temporal dimensions, respectively. Through a cross-attention mechanism, each query is constrained to attend only to historical features at the corresponding spatial location and to local features at the corresponding temporal position. Anomaly inference is conducted based on the similarity between the extracted global and local features of future frames and their corresponding predicted values. The overall architecture of the proposed self-supervised spatiotemporal feature relocalization framework is shown in Figure 2.

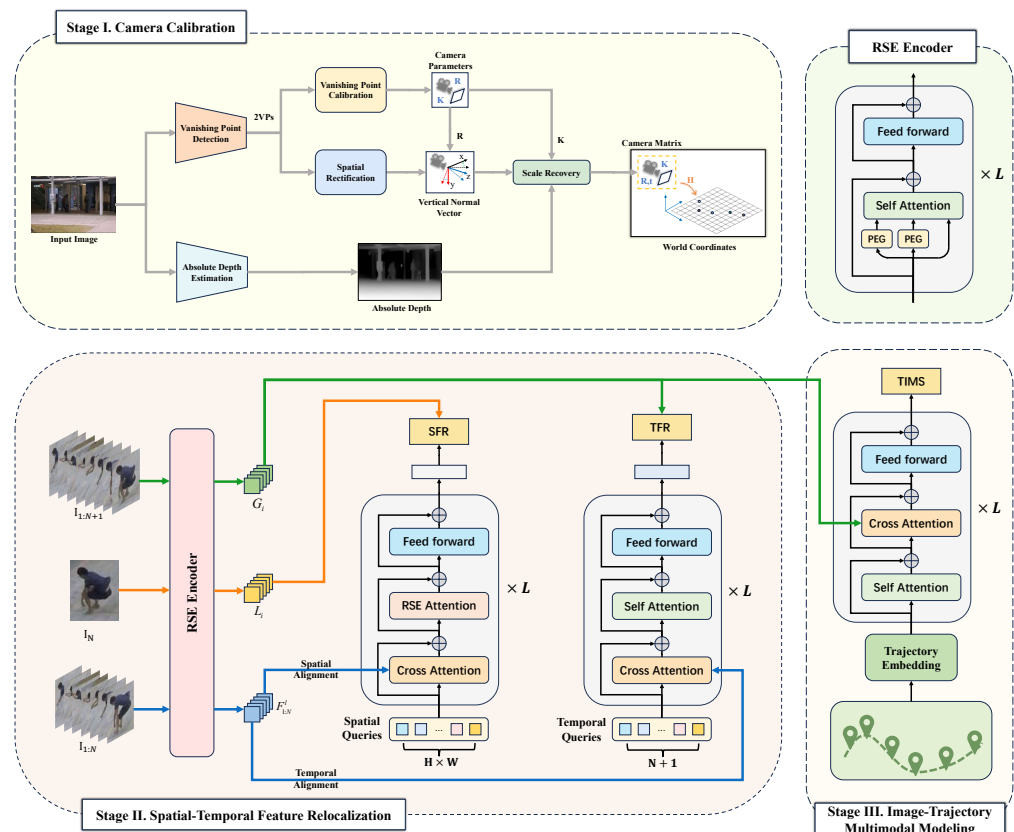


Figure 1. Overview of the proposed self-supervised video anomaly detection framework. The method consists of three stages: (I) spatial–temporal feature relocalization for discriminative representation learning, (II) camera calibration for recovering physical trajectories, and (III) image–trajectory multimodal modeling for enforcing consistency between pixel-space projections and real-world motion.

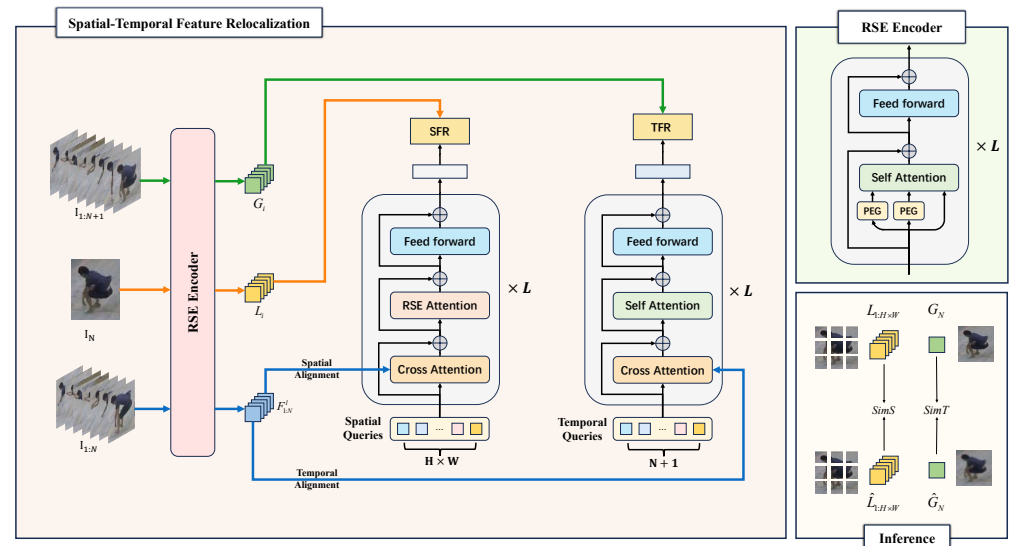


Figure 2. The overall architecture of the proposed self-supervised spatio-temporal feature relocalization framework for video anomaly detection. The RSE Encoder, constructed by stacking RSE attention layers and feed-forward networks, extracts feature targets to optimize the contrastive learning objective. While anomaly inference is achieved by calculating the similarity between the predicted and actual extracted spatio-temporal features of future frames.

3.1.1. Relative Spatial Encoding

To extract spatial–temporal information from video sequences and enable self-supervised feature relocalization, we adopt Vision Transformer as the backbone to model regional positional relationships and the temporal evolution of global features. To prevent training collapse caused by positional label leakage in self-supervised learning, we propose a Relative Spatial Encoding (RSE) module.

Existing studies predominantly rely on absolute positional encoding [25], learnable positional encoding [26], or rotary positional encoding [27] to facilitate global or local representation learning. However, both absolute and learnable positional encodings introduce explicit token-level absolute position priors to strengthen spatial relationship modeling, which can be inadvertently exploited in self-supervised spatial relocalization tasks, leading to positional label leakage and potential training collapse. Rotary positional encoding (RoPE) incorporates relative positional information by applying rotations to query and key vectors. Nevertheless, although absolute positions are not explicitly encoded, the rotation patterns remain consistent for identical pixel locations, which may still expose positional cues and thereby undermine the stability of self-supervised learning.

Inspired by CPVT [28], which introduces a Conditional Positional Encoding Generator (PEG) for modeling positional information based on local token neighborhoods, in this approach, the serialized one-dimensional token sequence is first reshaped back to its corresponding two-dimensional spatial feature map. A 2D convolution is then applied to capture local neighborhood interactions, from which positional encodings are generated according to the relative relationships among neighboring features. The generated positional encodings are subsequently added to the original token sequence. In addition, CvT [29] further demonstrates the feasibility of constructing positional encodings from relative token neighborhoods instead of relying on fixed positional encodings.

Following this idea, this work adopts a Conditional Positional Encoding Generator (PEG) to dynamically generate positional encodings associated with feature regions. The positional relationships are inferred from interactions among local feature neighborhoods rather than predefined spatial indices. To improve computational efficiency while preserving neighborhood modeling capability, depthwise separable convolution is employed to implement the conditional positional encoding generation process.

To prevent potential self-supervised spatial label leakage caused by the explicit introduction of positional encodings, inspired by rotary positional encoding [27], we apply the PEG module to the query and key vectors, while preventing the value vectors from incorporating relative positional information, thereby forming a relative spatial encoding attention mechanism. The formulation is given as follows:

$$RSE(Q, K, V) = softmax\left(\frac{(\varphi(Q))(\varphi(K))^T}{\sqrt{d_k}}\right)V \quad (1)$$

where $\varphi(\cdot)$ denotes the PEG module, Q , K , and V represent the query, key, and value vectors, respectively, $\sqrt{d_k}$ denotes the square root of the query feature dimension, and the $softmax(\cdot)$ function normalizes the similarity into a probability distribution.

Compared with Rotary Positional Encoding (RoPE), the RSE module constructed via PEG effectively prevents positional label leakage in self-supervised learning. The primary reason is that RoPE applies the same rotation-based encoding pattern to the same spatial positions across different samples. Consequently, the model tends to bypass the learning of meaningful visual features and instead relies on these fixed rotation patterns to complete the spatial relocalization task, leading to a failure in feature representation learning. In contrast, by leveraging the local receptive field and weight-sharing properties of convolutional neural networks, the PEG-based RSE module generates positional encodings solely based

on local feature neighborhoods. This ensures that the encoding process is data-dependent and localized, thereby eliminating the risk of spatial label leakage caused by identical encoding patterns at the same coordinates across different samples.

Based on the above Relative Spatial Encoding (RSE) module, the video sequence is encoded to obtain the local feature representations $F_{t,i}^l$ for each frame and the corresponding global feature representations F_t^g , where F denotes the features output by the RSE module, t represents the temporal index, and i denotes the spatial position index.

The spatio-temporal relocation module takes the local features $E = F_{1:N}^l$ from the previous N frames as input and constructs two sub-tasks, namely spatial relocation and temporal relocation. Specifically, the local features $L_i = F_{N+1,i}^l$ of frame $N + 1$ are used as the feature prediction targets for spatial relocation, while the global features $G_i = F_i^g$ from the previous $N + 1$ frames are used as the feature prediction targets for temporal relocation. In this manner, local spatial semantic consistency and global temporal semantic evolution are respectively constrained within a unified feature space.

3.1.2. Spatial Feature Relocalization Module

In this paper, the Spatial Relocalization Module aims to predict and relocalize the feature at each spatial position in frame $N + 1$ by leveraging the historical features of corresponding spatial locations from the preceding N frames. To avoid introducing absolute spatial priors that may lead to spatial label leakage, we employ a set of shared-parameter learnable query vectors, which do not encode any explicit spatial index information.

At each layer, the learnable query vectors are first used as queries, and a feature-space alignment strategy is adopted to aggregate local temporal semantic information from the corresponding spatial features of the preceding N frames via cross-attention. Subsequently, self-attention is applied after the Relative Spatial Encoding (RSE) module to model the discriminative relationships among features at different spatial positions.

Finally, the similarity between the predicted features and the extracted target features at each spatial position in frame $N + 1$ is computed as:

$$S_{i,j} = \frac{L_i \cdot \hat{L}_j}{\|L_i\| \|\hat{L}_j\|} \quad (2)$$

where $S_{i,j}$ denotes the spatial feature similarity, L_i represents the target feature of the i -th token, and $\hat{L}_j$ denotes the predicted feature of the j -th token. The feature at the corresponding spatial position is treated as a positive sample, while features at non-corresponding positions are regarded as negative samples. Based on this spatial matching strategy, a self-supervised contrastive learning objective is constructed using the InfoNCE loss as the optimization target of the Spatial Feature Relocalization (SFR) task:

$$L_S = -\frac{1}{n} \sum_{i=1}^n \log \frac{\exp(S_{i,i}/\tau)}{\sum_{j=1}^n \exp(S_{i,j}/\tau)} \quad (3)$$

where n denotes the total number of categories, which in this case corresponds to the number of spatial tokens, $\exp(\cdot)$ represents the exponential function, and τ is the temperature parameter. Through this objective, the model is encouraged to learn discriminative semantic representations rather than relying on pixel-level or feature-level reconstructability, by constructing a contrastive learning task between the predicted tokens and the corresponding ground-truth tokens of future frames, which pulls together predictions and ground truth at the same spatial location while pushing apart those from different locations, thereby guiding the model to focus on region-specific discriminative semantic features. During the

spatial relocation process, both cross-attention and self-attention follow the standard scaled dot-product attention formulation [25]. The detailed mathematical definition will be uniformly presented in the subsequent image–trajectory multimodal fusion module.

3.1.3. Temporal Feature Relocalization Module

The essence of the temporal relocalization module is to predict and localize the global feature $G_{1:N}$ of frame $N + 1$ by leveraging the global image information from the preceding N frames. To ensure that temporal modeling is conducted under consistent spatial semantic conditions, a query-guided temporal relocalization strategy is adopted. Specifically, a feature temporal alignment strategy and cross-attention are first employed to inject the overall spatial semantics of the current frame into the temporal queries of the preceding N frames, followed by self-attention along the temporal dimension to model temporal dependencies.

Subsequently, the predicted global feature of frame $N + 1$ is obtained, and its similarity with the target global features of each frame is computed:

$$T_i = \frac{G_i \cdot \hat{G}_{N+1}}{\|G_i\| \|\hat{G}_{N+1}\|} \quad (4)$$

where, T_i denotes the temporal feature similarity, G_i represents the target feature of the i -th token, and $\hat{G}_{N+1}$ denotes the predicted feature of future frame. Finally, the predicted future global feature and the corresponding real future feature are treated as a positive pair, while the predicted feature and the ground-truth global features of the preceding N frames are regarded as negative samples. Based on this temporal matching strategy, a self-supervised contrastive learning objective is constructed using the InfoNCE loss as the optimization target of the Temporal Feature Relocalization (TFR) task:

$$L_T = -\log \frac{\exp(T_{N+1}/\tau)}{\sum_{j=1}^n \exp(T_j/\tau)} \quad (5)$$

where n denotes the total number of categories, which in this case corresponds to the total number of temporal tokens, $\exp(\cdot)$ represents the exponential function, and τ is the temperature parameter. Through this objective, the model is encouraged to semantically approach the global feature of the future frame while being pushed away from the global features of the preceding N frames, thereby guiding the model to learn discriminative local tokens by pulling predictions closer to the ground truth at the same spatial location and pushing them away from those at different locations, allowing the network to focus on region-specific semantic discrimination rather than pixel-level reconstructability. The cross-attention and self-attention mechanisms used in the spatial relocalization module both follow the standard scaled dot-product attention formulation [25], and the unified mathematical definition of the attention operation will be provided in the subsequent image–trajectory multimodal fusion module.

3.2. Camera Calibration Model

The core objective of the camera calibration module is to accurately map object trajectories from the two-dimensional pixel space to the real-world coordinate system. To this end, the intrinsic and extrinsic parameters of the camera are estimated to recover the spatial geometric structure of the scene, thereby providing physically meaningful and high-precision trajectory representations for subsequent anomaly detection. To achieve this goal, we design a vanishing point detection and calibration module, as well as a depth-based geometric rectification and scale recovery module.

3.2.1. Camera Calibration Principles

Distinct from the calibration process that recovers spatial relationships via scale and distance based on the pinhole imaging principle, this section treats the camera calibration task as the computation of a homography matrix [30]. This matrix is utilized to project objects from a real-world plane onto the camera coordinate system. The calibration procedure, which maps pixel space to the physical ground plane, can be summarized as a process of calculating the camera homography matrix and performing the projection:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} r_1 & r_2 & t \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \quad (6)$$

where s is a scale factor; $\begin{bmatrix} u & v & 1 \end{bmatrix}^T$ and $\begin{bmatrix} X & Y & 1 \end{bmatrix}^T$ represent the 2D homogeneous coordinates in pixel space and physical space, respectively; r_i denotes the i -th column of the rotation matrix; and t is the translation vector. Essentially, performing camera calibration through this method centers on the process of calculating the homography matrix for the calibration task.

3.2.2. Vanishing Point Detection and Calibration

Vanishing points indicate the convergence locations of parallel lines in 3D space. We first employ a pre-trained deep learning-based method [22] to detect these points, which provide critical geometric cues regarding the camera perspective. Assuming the principal point is located at the image center, we interpret the top-3 detected vanishing points as corresponding to the vertical direction and two horizontal directions (i.e., width and depth extension). Specifically, we identify the two points aligned with the ground plane as the horizontal vanishing points. Subsequently, utilizing the geometric property of orthogonality—where the rays connecting the optical center to orthogonal vanishing points are perpendicular—we calculate the focal length, as formulated in Equation (7):

$$f = \sqrt{-(VP_1 - PP) \cdot (VP_2 - PP)} \quad (7)$$

where PP , VP_1 , and VP_2 denote the 2D coordinates of the principal point and the horizontal and vertical vanishing points, respectively. This process yields an initial estimate of the camera's intrinsic parameters, facilitating subsequent geometric correction and trajectory mapping.

Upon determining the focal length, the camera intrinsic matrix is estimated. To simplify the formulation, we assume zero principal point offset (the principal point coincides with the image center) and a unit aspect ratio with square pixels. The intrinsic matrix K is defined as:

$$K = \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8)$$

where f is the focal length derived from Equation (7), and (u_0, v_0) represent the coordinates of the principal point. Subsequently, we utilize the method proposed in [31] to compute the rotation matrix R , thereby refining the intrinsic and extrinsic parameters within the homography framework.

3.2.3. Spatial Rectification Module

Inspired by the work of DNet [32], we leverage depth estimation results to construct a point cloud representation in the camera coordinate system, achieving absolute scale recovery through 3D spatial structure modeling. However, DNet relies on the KITTI

dataset [33], which assumes zero pitch and roll angles when recovering the actual camera height. This leads to limited generalization in conventional scenarios where the pitch and roll angles are non-zero. In this section, we utilize the vanishing point calibration results to correct the plane normal vector within the camera coordinate system, ensuring robust adaptation to diverse perspectives.

While vanishing point calibration corresponds to the projection process of the world coordinate system, depth-based projection occurs within the camera coordinate system. To align these two representations, we incorporate image geometric cues to rectify the vertical normal vector in the camera coordinate system.

Specifically, the vertical normal vector V_W from the world coordinate system is transformed into the camera frame via the rotation matrix R calculated from vanishing point calibration. This process yields a rectified vertical normal vector V_I after image geometric semantic correction:

$$V_I = R^{-1} \cdot V_W \quad (9)$$

where V_W denotes the vertical normal vector in the world coordinate system, R is the rotation matrix derived from vanishing point calibration, and V_I represents the rectified normal vector in the camera coordinate system.

3.2.4. Scale Recovery Module

Compared to vanishing point detection, which primarily provides image-level geometric cues, depth estimation more effectively reflects the spatial information inherent in the scene, offering a complementary spatial understanding for camera calibration. In this section, we propose a depth-geometric scale recovery module. This module incorporates a pre-trained depth estimation model [34] to obtain absolute depth and utilizes the rectified vertical normal vector derived in the previous section to recover the absolute scale.

Specifically, we first leverage the depth estimation results and the calculated intrinsic matrix to project pixel coordinates into a point cloud representation within the camera coordinate system. This process can be formulated as follows:

$$D_t^{rel}(p_{i,j}) \cdot p_{i,j} = K \cdot P_{i,j} \quad (10)$$

where $p_{i,j}$ denotes the homogeneous pixel coordinates, $D_t^{rel}(p_{i,j})$ represents the absolute depth prediction at the corresponding position, K is the camera intrinsic matrix obtained from the vanishing point calibration stage, and $P_{i,j}$ is the projection of the pixel into the camera coordinate system using the depth. Through this scheme, this paper utilizes the vanishing point calibration results to supplement the camera intrinsics missing in the DNet [32] depth reprojection. Subsequently, the height of each point in the camera coordinate system relative to the camera optical center is calculated as:

$$h(P_{i,j}) = V_I \cdot \overrightarrow{OP_{i,j}} \quad (11)$$

where $h(P_{i,j})$ denotes the height at the corresponding pixel position, V_I is the vertical normal vector of the camera coordinate system, and $\overrightarrow{OP_{i,j}}$ represents the vector from the camera center to the projected point calculated following the method in DNet. This procedure effectively prevents errors caused by directly utilizing the vertical normal for height calculation.

Subsequently, the similarity between the 3D point cloud surface normals and the corrected camera vertical normal is calculated to filter ground points:

$$S = \left| \frac{V_I \cdot N(P_{i,j})}{\|V_I\| \|N(P_{i,j})\|} \right| \quad (12)$$

where V_I is the camera vertical normal, and $N(P_{i,j})$ is the surface normal estimated from the pixel neighborhood of $P_{i,j}$. Finally, the pixel positions corresponding to the top 50% similarity values are visualized and identified as ground points, as shown in Figure 3.

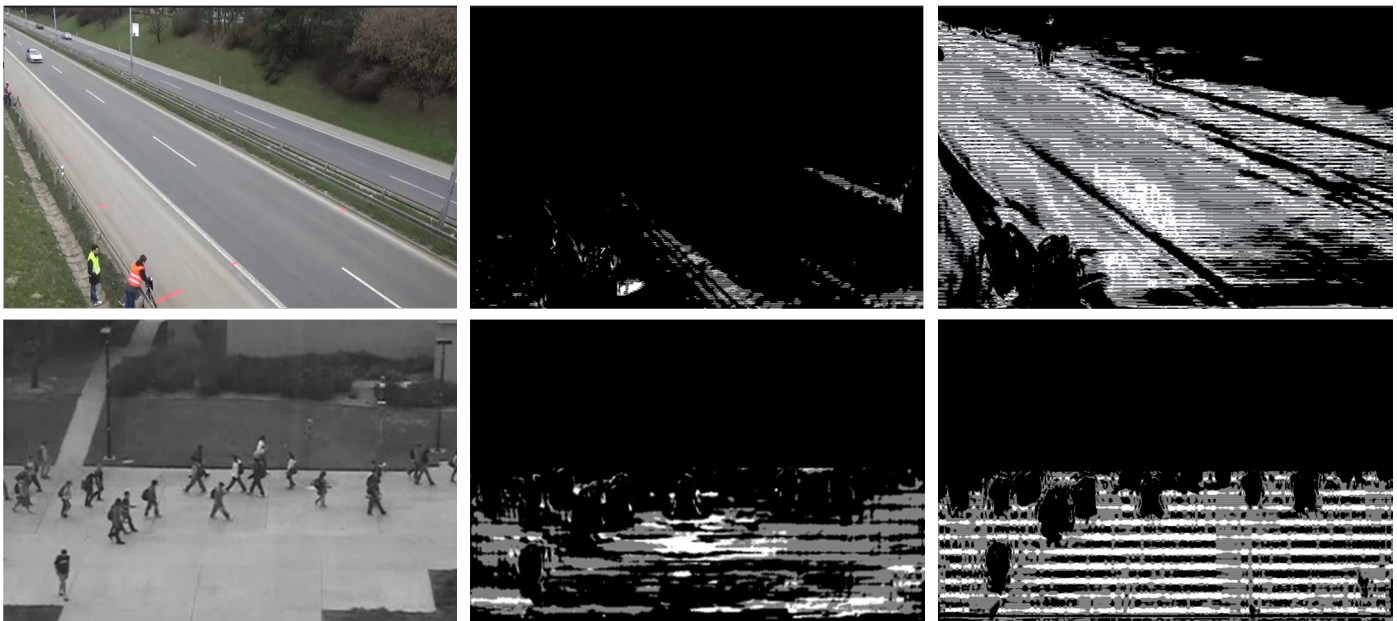


Figure 3. Visualization of ground point estimation results compared with DNet [32].

As illustrated in the figure, although the proposed method achieves more accurate ground point estimation compared to DNet, it still exhibits a bias toward planar estimation. Furthermore, points from non-flat regions, such as road bumps and depressions, are included in the computation, which interferes with the overall accuracy of the height estimation.

To address these issues, we utilize SAM [35] with “road” as a text prompt for further refinement of the ground points. The final height estimation is obtained by averaging the points whose surface normals fall within the top 50% of similarity to the vertical normal vector. Since the videos in video anomaly detection datasets are captured from fixed viewpoints, there is no need to establish a unified coordinate system for multiple cameras. Consequently, this work only utilizes the camera height to recover the translation matrix and refine Equation (6), thereby achieving the real-scale restoration of the calibration matrix to derive the homography matrix H .

3.3. Image–Trajectory Multimodal Modeling

Most existing approaches primarily rely on learning representations in the pixel space, overlooking the fact that objects simultaneously possess geometric projections in the physical world. The spatial–temporal relocalization-based self-supervised modeling paradigm focuses on capturing the spatial arrangement and temporal evolution of features within the pixel domain, yet it does not explicitly account for physical-space consistency. To address this limitation, we leverage the camera calibration module to recover real-world trajectories and further introduce a multimodal image–trajectory modeling module, which enhances the understanding of object behaviors from both physical-space geometry and pixel-space dynamics.

3.3.1. Reprojection of Physical Trajectories

In this section, the state-of-the-art YOLOv11 [36] architecture is implemented to perform robust object detection, which is further integrated with the ByteTrack [37] association

algorithm to extract motion trajectories within the pixel coordinate space. Subsequently, a homography-based calibration matrix is leveraged to project these pixel-level coordinates into the physical world coordinate system:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = H \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \quad (13)$$

where s denotes the scaling factor, and $\begin{bmatrix} u & v & 1 \end{bmatrix}^T$ and $\begin{bmatrix} X & Y & 1 \end{bmatrix}^T$ represent the 2D homogeneous coordinates in the pixel domain and the physical spatial domain, respectively. This mapping procedure effectively yields the absolute physical coordinates of the targets $P = \begin{bmatrix} X & Y \end{bmatrix}$; the reprojection of trajectories into the physical space using the calibration matrix is illustrated in Figure 4.

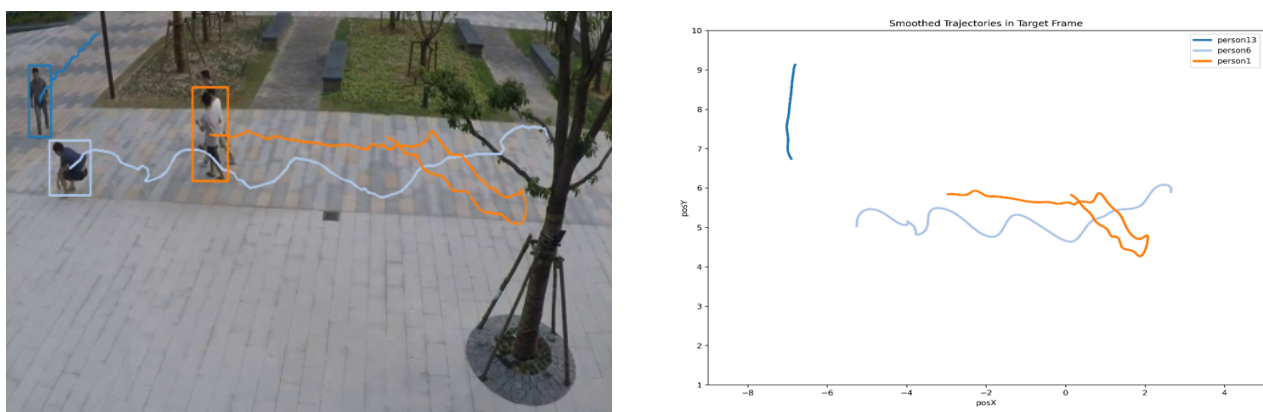


Figure 4. Visualization of trajectory projection from pixel space to world coordinates.

3.3.2. Trajectory Embedding

While most existing models normalize inputs and employ temporal learning through feature extraction backbones such as TCN or LSTM, this study explores the application of absolute positional encoding for trajectory embedding to preserve absolute spatial relationships:

$$TE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}}) \quad (14)$$

$$TE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}}) \quad (15)$$

where pos denotes the absolute spatial coordinates of the trajectory, i represents the index of the feature dimension, and d_{model} signifies the total dimensionality of the feature space. The positional results from the two-dimensional coordinates are subsequently concatenated along the feature channel dimension to generate the final trajectory embedding. Through this procedure, the absolute positions of the trajectory are effectively encoded into high-dimensional feature vectors.

3.3.3. Image–Trajectory Fusion Module

Subsequently, we construct an Image–Trajectory Fusion Module comprising a stack of self-attention, cross-attention, and feed-forward layers to jointly characterize the internal temporal dependencies of trajectory sequences and the cross-modal associations between trajectories and visual features. Specifically, the self-attention mechanism is first

implemented to model the trajectory features, thereby capturing long-range temporal dependencies between trajectory points. The attention computation is formally defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (16)$$

where Q , K , and V denote the query, key, and value vectors, respectively, and $\sqrt{d_k}$ represents the scaling factor derived from the square root of the feature dimension. In the self-attention stage, Q , K , and V are all derived from the trajectory embeddings. The cross-attention mechanism operates on a similar principle, wherein Q is sourced from the trajectory embeddings, while K and V represent the global visual features extracted from the respective frames of the image stream.

3.3.4. Image–Trajectory Multimodal Structuring Module

Upon obtaining the fused image–trajectory features, an Image–Trajectory Multimodal Structuring (ITMS) module is further developed to perform discriminative induction on the multimodal representations. This module introduces the image context to conditionally structure the trajectory features, ensuring that the image modality primarily provides semantic constraints and scene priors. Consequently, this prevents the discriminative process from being dominated by appearance-based information. Within this framework, discriminative supervision is applied exclusively to the trajectory-led fused representation, guiding the model toward a consistent induction of pixel-domain image features and physical-space trajectory features.

To mitigate the challenge of insufficient supervision signals in unsupervised video anomaly detection (VAD) tasks, the object categories derived from the tracking results (Section 3.3.1) are transformed into one-hot probability vectors to serve as class pseudo-labels. The image–trajectory behaviors are then regularized via a cross-entropy loss:

$$L_{CE} = -\sum_{i=1}^n y_i \log(\hat{y}_i) \quad (17)$$

where y denotes the pseudo-label generated by the aforementioned pre-trained visual classifier, y_i represents the predicted category probability, and n signifies the total number of classification categories.

3.4. Loss Function

During the training stage, to accommodate different spatio-temporal modeling objectives, independent optimization targets are constructed for the spatio-temporal relocalization module and the image–trajectory multimodal induction module, respectively. Separate training strategies are adopted to avoid mutual interference between different objectives in multi-task joint optimization.

The spatio-temporal relocalization module consists of two sub-tasks, namely spatial relocalization and temporal relocalization, which constrain the spatial semantic consistency of local tokens and the temporal evolution consistency of global tokens, respectively. In this way, the model is encouraged to learn stable and discriminative normal spatio-temporal structures. The overall loss function is defined as:

$$L_R = \frac{L_S + L_T}{2} \quad (18)$$

where L_S and L_T denote the spatial relocalization loss and the temporal relocalization loss, respectively, whose detailed formulations have been provided in Section 2.1.

During training, the proposed framework performs parallel optimization of the spatio-temporal relocalization loss L_R and the image–trajectory induction loss L_{CE} , and updates

the gradients of the dual-branch network separately, thereby improving overall training stability and enhancing the generalization capability of the model.

3.5. Anomaly Inference

During the inference stage, to prevent relative differences among features from interfering with anomaly judgment, we do not adopt probability outputs normalized by softmax. Instead, we directly compute the normality scores based on the cosine similarity results S_s and S_t produced by the spatio-temporal relocation task:

$$S^t = \min_i S_{i,i}^t \quad (19)$$

$$T^t = T_i^t \quad (20)$$

where S^t denotes the spatial normality score at time t , and $S_{i,i}^t$ represents the cosine similarity between the future feature at the corresponding spatial location and its estimated feature; T^t denotes the temporal normality score at time t , and T^{N+1} represents the cosine similarity between the global feature of the future frame and its estimated feature. The former takes the minimum similarity between the ground-truth and predicted features at corresponding spatial locations as the normality score, while the latter directly uses the similarity between the global feature of the future frame and its estimation as the normality score.

For the computation of the normality score in the image–trajectory multimodal induction module, considering the invisibility and non-enumerability of anomalies under unsupervised training, we propose to utilize the maximum confidence class deviation as the normality scoring criterion:

$$id = \arg \max_i (\hat{y}_i) \quad (21)$$

$$SC^t = \text{abs} \left(\max_i (\hat{y}_i) - y_{id} \right) \quad (22)$$

where id denotes the class with the highest confidence, $\arg \max(\cdot)$ indicates the selection of the category with the highest predicted probability, and $\hat{y}_i$ represents the predicted label; SC^t denotes the multimodal normality score at time t , $\text{abs}(\cdot)$ denotes the absolute value function, and y_{id} represents the index of the category corresponding to the highest-confidence class.

Finally, the normalized anomaly score is computed for each branch:

$$a^t = \frac{p^t - \min_t p^t}{\max_t p^t - \min_t p^t} \quad (23)$$

where a^t denotes the anomaly score of the t -th frame, and p^t represents the anomaly score produced by each stream before normalization. Accordingly, the anomaly scores from spatial relocation, temporal relocation, and image–trajectory multimodal induction are normalized separately, yielding A_i^S , A_i^T , and A_i^M , respectively.

Considering the absence of anomaly labels during the self-supervised training stage, in order to preserve the discriminative characteristics of spatial, temporal, and multimodal anomaly cues while obtaining a unified anomaly evaluation metric, we first take the maximum value between the normalized spatial and temporal anomaly scores within the spatio-temporal relocation branch to enhance sensitivity to prominent abnormal patterns. The resulting score is then averaged with the multimodal anomaly score to obtain the final anomaly score:

$$A_i = \frac{\min(A_i^S, A_i^T) + A_i^M}{2} \quad (24)$$

where A_i denotes the final anomaly score of the t -th frame, and A_t^S , A_t^T , and A_t^M represent the normalized anomaly scores of spatial relocalization, temporal relocalization, and image–trajectory multimodal induction, respectively.

4. Experiments and Analysis

4.1. Experimental Setup

4.1.1. Datasets

To validate the effectiveness and robustness of the proposed method, experiments are conducted on three widely used benchmark datasets for video anomaly detection: UCSD Ped2 [38], CUHK Avenue [39], and ShanghaiTech Campus [40]. The UCSD Pedestrian dataset consists of two subsets (Ped1 and Ped2) captured by static cameras from different viewpoints; in this work, we adopt the Ped2 subset, which is recorded from a relatively top-down perspective. UCSD Ped2 contains 16 training videos and 12 testing videos, with a total of 21 abnormal events, primarily involving anomalies in appearance and motion such as bicycles and vehicles appearing in pedestrian areas. The CUHK Avenue dataset is collected using an outdoor surveillance camera monitoring a campus walkway, focusing mainly on pedestrian activities; it includes 16 training videos and 21 testing videos with 77 abnormal events, most of which are motion-related anomalies such as running, throwing, and sudden abnormal behaviors. The ShanghaiTech Campus dataset is a large-scale and more challenging benchmark captured by multiple cameras across diverse campus scenes, where even videos of the same scene may exhibit different viewpoints, leading to significant perspective distortion and scale variation. It comprises 330 training videos and 107 testing videos with 158 abnormal events, covering both abnormal appearances (e.g., bicycles, skateboards, and cars) and abnormal motions (e.g., falling, crossing, and chasing).

4.1.2. Evaluation Metrics and Implementation Details

All experiments are conducted on a workstation equipped with an NVIDIA GeForce RTX 1080Ti GPU, 16 GB RAM, running Windows 10, and configured with CUDA 10.2 for GPU acceleration. The proposed framework is implemented in Python 3.7 using PyTorch 1.8. The learning rate is set to 1×10^{-4} , and the model parameters are optimized using the Adam optimizer to ensure stable and efficient convergence. Notably, the temperature parameter τ in the InfoNCE loss is configured as a learnable parameter with an initial value of 0.07, allowing the model to dynamically calibrate the concentration of feature distributions across different scenarios. For quantitative evaluation, we adopt the micro-averaged Area Under the Receiver Operating Characteristic curve (AUROC) [41] as the frame-level evaluation metric. Specifically, the predicted anomaly score for each frame is compared against the corresponding ground-truth annotation to compute the AUROC, which measures the trade-off between true positive rate and false positive rate across different thresholds. Micro-averaging aggregates all frame-level predictions over the entire dataset, providing a comprehensive and fair assessment of overall detection performance.

In addition, to report the computational cost of the proposed framework, we provide the training time under the aforementioned hardware configuration. The model is trained for 80, 40, and 20 epochs on the UCSD Ped2, Avenue, and ShanghaiTech datasets, respectively. The corresponding training times are approximately 1.5 h, 8 h, and 25 h.

We further compare the inference efficiency with several representative object-level video anomaly detection methods. HF²-VAD [42], Jigsaw-VAD [16], and AED [41] achieve approximately 8 FPS, 23 FPS, and 19 FPS, respectively, whereas the proposed method achieves 27 FPS. Although the proposed framework incorporates both image and trajectory modalities, the computational overhead remains moderate since the self-supervised spatio-temporal repositioning task enables the model to learn fine-grained representations

without relying on computationally intensive 3D convolutions or pixel-level reconstruction decoders. Consequently, the proposed method maintains competitive frame-level AUC performance while achieving superior inference efficiency.

4.2. Comparison with State-of-the-Art Methods

To verify the effectiveness and competitiveness of the proposed framework for the video anomaly detection (VAD) task, we conduct comprehensive comparisons with several recent state-of-the-art (SOTA) methods spanning different modeling paradigms, including prediction-based, reconstruction-based, generative, diffusion-based, and so on. And the compared methods include Frame-Pred [43], SSAGAN [44], DLAN-AC [3], PseudoBound [45], DSS-Net [46], USTN-DSC [47], AED-MAE [48], MGAN-CL [6], VADiffusion [49], DS-AND [50], MA-PDM [51], and MemoryOut [52]. These methods represent diverse technical routes and recent advances in VAD, providing a comprehensive benchmark to evaluate the robustness, generalization ability, and practical effectiveness of our approach.

As summarized in Table 1, our proposed method achieves frame-level AUCs of 98.9% and 83.5% on the UCSD Ped2 and ShanghaiTech Campus (STC) datasets, outperforming the second-best methods by 0.2% and 0.5%, respectively.

The performance gain is primarily attributed to two factors: First, for the Ped2 dataset, the feature-level discriminative learning effectively suppresses the negative impact of background noise and scale fluctuations. Second, for the STC dataset, which is characterized by multi-scene and multi-view environments, the integration of pixel-level feature learning with absolute trajectory understanding successfully overcomes drastic scale variations, significantly enhancing the model's robustness in complex scenarios.

Furthermore, our method trails the state-of-the-art (SOTA) MemoryOut on the CUHK Avenue dataset by a marginal 0.1%. This slight discrepancy likely stems from the high similarity between specific abnormal appearances and the granularity noise induced by normal scale changes in this dataset, suggesting a need for even finer-grained pixel-level analysis.

Table 1. Frame-level AUROC (%) comparison with state-of-the-art methods on UCSD Ped2, CUHK Avenue, and ShanghaiTech Campus (STC) datasets.

Method	Ped2	Avenue	STC
Frame-Pred	95.4	85.1	72.8
SSAGAN	97.6	88.9	74.3
DLAN-AC	97.6	89.9	74.7
PseudoBound	98.4	87.1	73.7
DSS-Net	97.2	90.6	75.5
USTN-DSC	98.1	89.9	73.8
AED-MAE	95.4	91.3	79.1
MGAN-CL	96.5	87.1	73.6
VADiffusion	98.2	87.2	71.7
DS-AND	98.7	87.4	71.6
MA-PDM	98.6	91.3	79.2
MemoryOut	98.4	91.6	83.0
Ours	98.9	91.5	83.5

In summary, these results demonstrate that our self-supervised spatial-temporal repositioning contrastive learning framework serves as a superior alternative to traditional pixel-level reconstruction or prediction tasks. By leveraging the multimodal image-trajectory induction module, the model significantly bolsters its comprehension of pixel-level appearance and temporal motion, while effectively utilizing physical spatial information to reinforce the understanding of object behaviors.

4.3. Ablation Studies

4.3.1. Effectiveness of Relative Spatial Encoding Model

To verify the effectiveness of the proposed Relative Spatial Encoding (RSE) module, we compare our method with three prominent position encoding techniques: Absolute Position Encoding [25], Learnable Position Encoding [26], and Rotary Position Encoding [27]. The experimental results are summarized in Table 2.

Table 2. Ablation results of the Relative Spatial Encoding module.

Method	Ped2	Avenue	STC
Absolute Position Encoding	92.8	86.1	75.3
Learnable Position Encoding	89.5	83.7	73.1
Rotary Position Encoding	94.9	89.6	77.7
Ours	98.9	91.5	83.5

The results indicate that assigning learnable position encodings to each token is more prone to spatial label leakage than absolute position encoding. While Rotary Position Encoding mitigates this issue to some extent, it still leads to a degradation in feature learning when the rotation patterns of the predicted and future features remain identical. In contrast, our method leverages the feature neighborhood for position encoding, effectively preventing spatial feature leakage inherent in fixed, learnable, or rotary position encodings.

4.3.2. Effectiveness of Self-Supervised Feature Relocation Task

To evaluate the effectiveness of the proposed Self-supervised Feature Repositioning (SFR) task, we compare three modeling paradigms designed to enhance spatio-temporal discriminative representations. These paradigms guide the model to focus on key spatio-temporal semantic information through distinct mapping logics: (1) utilizing a Multi-Layer Perceptron (MLP) for fundamental feature space projection; (2) utilizing a Masked Cross-Attention mechanism to simultaneously aggregate spatio-temporal contexts; and (3) extracting spatial-temporal relationships via the proposed query-separation mechanism. The experimental results are summarized in Table 3.

Table 3. Ablation results of the feature repositioning decoding module (Frame-level AUC %).

Method	Ped2	Avenue	STC
MLP	90.7	85.3	73.5
Masked Cross-Attention	96.3	90.1	79.4
Ours	98.9	91.5	83.5

As shown in Table 3, the proposed method significantly outperforms both the MLP and MCA baselines in terms of discriminative performance. The underlying reasons for these performance gaps can be summarized as follows:

The MLP paradigm is susceptible to “identity shortcuts” that trigger task collapse, while the Masked Cross-Attention paradigm suffers from feature entanglement due to simultaneous spatial-temporal encoding. In contrast, our method employs a query-separation mechanism to achieve explicit decoupling. This effectively mitigates information mixture and enforces discriminative feature extraction, thereby enhancing robustness against complex anomalies.

4.3.3. Effectiveness of Calibrated Trajectory Modeling

To evaluate the effectiveness of the proposed camera calibration and multi-modal image-trajectory induction modules, we conduct a comparative study across

five configurations: (1) Image-only Spatio-temporal Repositioning; (2) Single-modal Image Induction; (3) Pixel Trajectory Embedding; (4) Vanishing Point (VP)-based Trajectory Embedding; (5) Vanishing Point (VP)-Perturbed Trajectory Embedding; and (6) our proposed Calibrated Trajectory Embedding. The results are summarized in Table 4.

Table 4. Ablation analysis of calibrated trajectory modeling (Frame-level AUC %).

Method	Ped2	Avenue	STC
Image-only Repositioning	97.5	90.7	78.6
Single-modal Image Induction	97.2	89.8	78.1
Pixel Trajectory Embedding	98.3	91.1	80.4
VP-based Trajectory Embedding	97.3	90.4	81.2
VP-Perturbed Trajectory Embedding	92.7	85.3	77.1
Ours	98.9	91.5	83.5

As indicated in Table 4, our method consistently outperforms baselines that rely solely on image features or unrecovered trajectories. This validates the effectiveness of incorporating trajectory flows and restoring them to the world coordinate system for video anomaly detection.

It is worth noting that the VP estimation process primarily rectifies the camera's rotational deviation (perspective). However, in the absence of absolute scale recovery, discrepancies in camera height and focal length across different video scenes remain unaddressed, leading to scale inconsistency. Consequently, raw pixel trajectories may exhibit better local scale consistency within a fixed camera view than partially calibrated VP trajectories. Notably, when the vanishing point is subjected to severe perturbation (VP-Perturbed), the model performance drops significantly (e.g., from 98.9% to 92.7% on Ped2). This sharp decline underscores that incorrect geometric priors introduce misleading motion bias, which severely degrades the alignment between visual features and physical dynamics. In contrast, our method achieves unified physical scale normalization through geometric constraints. By mapping heterogeneous pixel displacements into motion features with explicit physical meaning, our approach significantly enhances the model's discriminative capability for complex anomalous events.

4.3.4. Effectiveness of Multi-Stream Fusion Strategies

To investigate the impact of score aggregation on anomaly detection, we evaluate six fusion paradigms by alternating the average and minimum operators across different branch levels. This study aims to identify the optimal balance between anomaly sensitivity and semantic stability by addressing the discrepancies between spatio-temporal streams (S, T) and the multimodal branch (M). The experimental results are summarized in Table 5.

Table 5. Ablation results of multi-stream fusion strategies (Frame-level AUC %).

Fusion Strategy	Ped2	Avenue	STC
$\text{avg}(S, T)$	95.4	87.2	78.5
$\text{min}(S, T)$	94.1	85.9	76.8
$\text{avg}(S, T, M)$	97.2	89.8	81.1
$\text{min}(S, T, M)$	96.1	88.5	79.8
$\text{avg}(\text{avg}(S, T), M)$	97.8	90.3	81.9
Ours	98.9	91.5	83.5

As illustrated in Table 5, our hierarchical strategy outperforms all baselines. The rationale for selecting the minimum operator at the spatio-temporal level is that an anomaly

in either the spatial or temporal domain should be sufficient to trigger an alert. To mitigate noise sensitivity, the multimodal (M) branch is subsequently integrated via the average operator; this dual-stage mechanism ensures the retention of discriminative anomaly cues while maintaining global semantic robustness.

4.3.5. Sensitivity Analysis of Temporal Context Length N

To evaluate the model's adaptability to different motion speeds, we compare three configurations ($N = 5, 7, 9$) in Table 6. Results show that shorter windows ($N = 5$) are highly sensitive to instantaneous, high-speed anomalies, whereas longer windows ($N = 9$) are better suited for persistent, low-speed patterns. Our selection of $N = 7$ provides the optimal balance, ensuring robust performance across diverse motion intensities.

Table 6. Impact of temporal context length N on frame-level AUC (%).

Context Length N	Ped2	Avenue	STC
$N = 5$	98.1	89.2	81.2
$N = 7$	98.9	91.5	83.5
$N = 9$	98.4	90.1	82.6

5. Discussion

In this paper, we revisit the problem of video anomaly detection from the perspective of feature modeling. To address the limitations of traditional reconstruction-based methods—specifically their sensitivity to background noise and scale variations—we propose a self-supervised learning framework centered on Spatio-temporal Feature Repositioning. By leveraging feature-level repositioning, the model actively aligns critical spatio-temporal semantic cues while suppressing interference from irrelevant background and scale variations. This mechanism significantly enhances the perception of both local appearance anomalies and global temporal irregularities in behavior. Furthermore, by incorporating physical motion information derived from camera calibration, we achieve a synergistic modeling of pixel-level features and trajectory embeddings, which effectively bolsters the representation of complex anomalous behaviors. Extensive experiments on multiple public datasets demonstrate the superior detection performance and stability of our method, validating the effectiveness of the proposed modeling paradigm.

Despite these contributions, the proposed method still has certain limitations. First, the camera calibration module is designed to restore real-world motion within a 2D ground plane, which presents an inherent deficiency in interpreting complex 3D spatial dynamics. Moreover, this calibration relies on highly simplified pinhole camera model assumptions, such as zero principal point offset, unit aspect ratio, and no lens distortion. In practice, obtaining precise intrinsic parameters fundamentally requires physical access to the specific camera, making accurate online self-calibration solely from video streams highly challenging. Given these practical constraints, omitting complex parameters during self-calibration has become a widely accepted standard practice in the field. This practice is further necessitated by the absence of intrinsic camera parameters in standard video anomaly detection datasets.

Second, the reliance on off-the-shelf object detection and tracking models for object-level modeling introduces a dependency bottleneck; specifically, missed detections in extreme environments characterized by complex optical imaging degradations (e.g., low-light noise, severe illumination changes, motion blur, and atmospheric scattering), low-confidence scenarios, or the presence of open-set objects can lead to omission of feature representations for these specific objects.

To address these challenges, future work will focus on constructing a unified appearance–trajectory–semantic perception framework. We aim to extend physical space recovery and motion analysis from the 2D plane to 3D space to capture fine-grained motion patterns. Simultaneously, we will introduce high-level semantic features to enhance the model’s generalized understanding of anomalous behaviors. Ultimately, we seek to achieve consistent modeling and unified detection across pixel-level appearance, real-world physical trajectories, and high-level motion semantics, thereby significantly improving robustness in complex real-world surveillance scenarios.

Furthermore, to specifically address the practical challenges of optical imaging degradation (e.g., low-light noise, motion blur, and atmospheric scattering), we plan to explore frequency-domain analysis to effectively isolate and mitigate such complex degradations. To overcome the scarcity of degraded training samples, we intend to employ deep generative models to synthesize extreme imaging conditions for robust data augmentation. To ensure rigorous evaluation, we aim to construct and release a novel, purely real-world video anomaly detection benchmark dataset specifically capturing these severe optical degradations. By bridging the gap between diverse synthetic training domains and real-world extreme evaluation scenarios, we expect to systematically enhance the model’s environmental robustness and propel future research toward reliable open-world surveillance.

In addition, inspired by recent advances in multimodal video understanding [53], future research will investigate the incorporation of the audio modality to further enrich environmental perception. Compared with purely visual signals, audio cues can provide complementary information about object interactions and surrounding events, which may not be fully observable from visual observations alone. Integrating audio–visual representations into the proposed perception framework is expected to facilitate a more comprehensive characterization of anomalous events and further enhance detection robustness in complex real-world surveillance environments.

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